*In Exercises 5 and 6, which of the given subsets of $R^3$ are subspaces?*

**5.** The set of all vectors of the form

**(a)** $\begin{bmatrix} a \\ b \\ 1 \end{bmatrix}$ **(b)** $\begin{bmatrix} a \\ b \\ a+2b \end{bmatrix}$ **(c)** $\begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix}$

**(d)** $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$, where $a + 2b - c = 0$

**6.** The set of all vectors of the form

**(a)** $\begin{bmatrix} a \\ b \\ 0 \end{bmatrix}$ **(b)** $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$, where $a > 0$

**(c)** $\begin{bmatrix} a \\ a \\ c \end{bmatrix}$ **(d)** $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$, where $2a - b + c = 1$

*In Exercises 7 and 8, which of the given subsets of $R_4$ are subspaces?*

**7.** **(a)** $\begin{bmatrix} a & b & c & d \end{bmatrix}$, where $a - b = 2$

**(b)** $\begin{bmatrix} a & b & c & d \end{bmatrix}$, where $c = a + 2b$ and $d = a - 3b$

**(c)** $\begin{bmatrix} a & b & c & d \end{bmatrix}$, where $a = 0$ and $b = -d$

**8.** **(a)** $\begin{bmatrix} a & b & c & d \end{bmatrix}$, where $a = b = 0$

**(b)** $\begin{bmatrix} a & b & c & d \end{bmatrix}$, where $a = 1, b = 0$, and $a + d = 1$

**(c)** $\begin{bmatrix} a & b & c & d \end{bmatrix}$, where $a > 0$ and $b < 0$

Solution:

2. Yes.

4. No.

6. (a) and (c).

8. (a).

**12.** Let $W$ be the set of all $3 \times 3$ matrices of the form

$$\begin{bmatrix} a & 0 & b \\ 0 & c & 0 \\ d & 0 & e \end{bmatrix}.$$

Show that $W$ is a subspace of $M_{33}$.

12. (a) Let

$$A = \begin{bmatrix} a_1 & 0 & b_1 \\ 0 & c_1 & 0 \\ d_1 & 0 & e_1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} a_2 & 0 & b_2 \\ 0 & c_2 & 0 \\ d_2 & 0 & e_2 \end{bmatrix}$$

be any vectors in $W$. Then

$$A + B = \begin{bmatrix} a_1 + a_2 & 0 & b_1 + b_2 \\ 0 & c_1 + c_2 & 0 \\ d_1 + d_2 & 0 & e_1 + e_2 \end{bmatrix}$$

is in $W$. Moreover, if $k$ is a scalar, then

$$kA = \begin{bmatrix} ka_1 & 0 & kb_1 \\ 0 & kc_1 & 0 \\ kd_1 & 0 & ke_1 \end{bmatrix}$$

is in $W$. Hence, $W$ is a subspace of $M_{33}$.

**14.** Let $W$ be the set of all $2 \times 2$ matrices

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

such that $A\mathbf{z} = \mathbf{0}$, where $\mathbf{z} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Is $W$ a subspace of $M_{22}$? Explain.

14. We have

$$A\mathbf{z} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} a+b \\ c+d \end{bmatrix},$$

so $A$ is in $W$ if and only if $a+b=0$ and $c+d=0$. Thus, $W$ consists of all matrices of the form

$$\begin{bmatrix} a & -a \\ c & -c \end{bmatrix}.$$

Now if

$$A_1 = \begin{bmatrix} a_1 & -a_1 \\ c_1 & -c_1 \end{bmatrix} \quad \text{and} \quad A_2 = \begin{bmatrix} a_2 & -a_2 \\ c_2 & -c_2 \end{bmatrix}$$

are in $W$, then

$$A_1 + A_2 = \begin{bmatrix} a_1 & -a_1 \\ c_1 & -c_1 \end{bmatrix} + \begin{bmatrix} a_2 & -a_2 \\ c_2 & -c_2 \end{bmatrix} = \begin{bmatrix} a_1+a_2 & -(a_1+a_2) \\ c_1+c_2 & -(c_1+c_2) \end{bmatrix}$$

is in $W$. Moreover, if $k$ is a scalar, then

$$kA_1 = k \begin{bmatrix} a_1 & -a_1 \\ c_1 & -c_1 \end{bmatrix} = \begin{bmatrix} ka_1 & -(ka_1) \\ kc_1 & -(kc_1) \end{bmatrix}$$

is in $W$. Alternatively, we can observe that every vector in $W$ can be written as

$$\begin{bmatrix} a & -a \\ c & -c \end{bmatrix} = a \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix},$$

so $W$ consists of all linear combinations of two fixed vectors in $M_{22}$. Hence, $W$ is a subspace of $M_{22}$.